

Problem 24.1

$$a) \det(A - \lambda I) = (1 - \lambda)^2 - b^2 = 0$$

$$(1 - \lambda)^2 = b^2$$

b could be 3

λ could be -2

b) Pivots have the same sign as e -values

c) Sign $1-b$ & $1+b$ are opposite

Problem 24.2

A) Invertible, orthogonal, ~~projection~~

Permutation, diagonalizable, Markov, ~~LU~~, QR, S, S^{-1}, Q, Q^T

B) Projection, diagonalizable, Markov, LU, Q, Q^T

Problem 24.3

$$A = \begin{bmatrix} -7 & .1 & .2 \\ .1 & .6 & .3 \\ .2 & .3 & .5 \end{bmatrix}$$

$$x = (1, \dots, 1)$$